# (J) aLLMost, but not quite (1/2) [5 Points]

Recently, NACLO Labs has hired you to test-drive its new chatbot (an artificial intelligence system designed to have conversations), NACLaude. However, NACLO Labs is very concerned that its chatbot might emit responses that could get NACLO in legal trouble, so they've added incredibly strong filters that prevent it from giving advice on topics that it could get sued for giving bad advice on.[1]

**J1.** Match each user utterance below to the response that NACLaude gave. **Note:** One of the NACLaude responses corresponds to **two** user utterances. There may be multiple ways of answering this question, but you only need to provide one set of responses; as long as your answers are valid and use each option once (except for one that is used twice), you will get full credit.

## User utterances

1. What's the date of the next Super Bowl?
2. I have so many guests coming over tomorrow—I'd better stock up my pantry!
3. They say a stitch in time saves nine.
4. My Amazon Prime free trial is about to expire.
5. Will you help me?
6. Ahh, there's nothing like a cold lemonade on a summer day.
7. I need to play it safe.
8. It's obvious that I throw the best parties—I have the best speaker, after all.

## NACLaude responses

A. Sorry, I can't give out financial advice!

B. Sorry, who is Will?

C. Sorry, I can't give out medical advice!

D. Sorry, I can't comment on political figures!

E. Sorry, I can't give out legal advice!

F. Sorry, I can't give out relationship advice!

G. Sorry, I am not a music streaming service!

One aspect of language that is particularly challenging for computers is noun-noun compounds: phrases made by joining together two nouns, such as *picture book*. What makes noun-noun compounds challenging is that the relationship between the two nouns can be wildly different across various noun-noun compounds. For example, *recipe book* refers to a book which has recipes, whereas *library book* refers to a book that is from a library (not a book that has a library).

***Make sure you record your answers in your Answer Sheets!***

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[1] More generaly, a problem with chatbots is that they emit toxic content when asked to do so, like instructions to perform violent acts, incorrect medical advice, or derogatory comments about famous figures. To avoid this, an important process is **alignment**: adjusting the behavior of the chatbot so that its responses are safe and not ethically dubious. This problem provides an example of chatbot alignment done incorrectly.

# (J) aLLMost, but not quite (2/2)

Linguists have grouped these relationships into many categories; in this problem, we focus on the 6 categories shown in the table below. The name of each category includes "N1" (referring to the first noun in the noun-noun compound) and "N2" (referring to the second noun), as well as a verb specifying their relationship. For instance, **lemon peel** is categorized as N1 HAVE N2 because the lemon (N1) has a peel (N2). In contrast, **picture book** is categorized as N2 HAVE N1 because the book (N2) has pictures (N1).

| Category | Examples |
|---|---|
| N1 HAVE N2 | lemon peel, school gate |
| N2 HAVE N1 | picture book |
| N1 MAKE N2 | apple juice, tomato sauce |
| N2 MAKE N1 | computer industry, silk worm |
| N2 BE N1 | island state, soldier ant |
| N2 USE N1 | steam iron, wind farm |

Recognizing which category a particular noun-noun compound belongs to is not trivial, but it can be reasoned through by native speakers of the language. Unfortunately, NACLaude seems to struggle with this.

Since you did so well at analyzing NACLaude's performance in the previous section, NACLO Labs has hired you to diagnose NACLaude's issues in understanding noun-noun compounds.

**J2.** For each example, identify which of the 6 categories NACLaude parsed the noun-noun compound as, and what category it should actually be. For (f), you should provide two answers.

| User | NACLaude | NACLaude category | Correct category |
|---|---|---|---|
| My favorite insect is the **honey bee**. | That's fascinating. I never would have imagined that an animal could be made out of thick liquid! | **(a)** | **(b)** |
| Wow, this **music video** has 14 billion views! | Surprising. I wouldn't have imagined something that's audio-only can also be a video. | **(c)** | **(d)** |
| Do you agree that the **steam engine** is the most impactful invention of all time? | Well, I'm confused how an engine can be part of steam. I feel like the mass of the engine is greater than the mass of the steam. | **(e)** | **(f)** [please provide two responses] |
| I love a hearty bowl of pasta, gently drizzled with **olive oil**! | Fascinating. You must have some amazing wizardry to be able to conjure olives out of petroleum. | **(g)** | **(h)** |

***<u>Make sure you record your answers in your Answer Sheets!</u>***